



Authorship transparency and equity in otolaryngology and maxillofacial surgery: current practices and the potential impact of large language models

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Received: 21 August 2024 / Accepted: 9 December 2024 / Published online: 19 December 2024
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Abstract

Purpose The determination of authorship order in scientific publications is critical for researcher recognition and career progression. According to a recent survey's meta-analysis, the prevalence of Honorary Authorship reaches 20% of authors. This study evaluates the adoption and proficiency of authorship guidelines in maxillofacial surgery and otolaryngology.

Methods On March 16, 2024, a search of Scopus-indexed journals identified relevant journals in from which recent articles were sampled using a stratified random method. The author contribution alignment with actual authorship was evaluated through CalculAuthor tool and a detailed process involving large language models (LLM) and manual checks, employing statistical tests to evaluate the adoption of author contribution guidelines and agreement scores across different fields and journal quartiles.

Results A review of 21 Scopus-indexed journals revealed that 8 adhered to the CRediT system, 7 to other systems, and 6 had no guidelines. From 147 randomly selected manuscripts, 57 reported contributions (38.8%). The mean agreement score was 65.3% (SD = 26.8), without significant differences in journal quartiles or disciplines.

Conclusions This study highlights the moderate to low adoption of standardized contribution systems in maxillofacial and otolaryngology journals, with a fair level of agreement between reported contributions and actual authorship. Future research should focus on larger, more diverse samples and develop prospective implementations to ensure accurate and equitable authorship credit in the context of the Artificial Intelligence's scientific writing era.

Keywords Authorship · CalculAuthor · CRediT · Maxillofacial surgery · Otolaryngology

Introduction

The determination of authorship order in scientific publications is a critical issue that directly impacts the recognition and career progression of researchers [1–3]. In medical research, the traditional approach to authorship attribution has often been criticized for lacking transparency and objectivity, leading to disputes and dissatisfaction among co-authors [4–6]. Many guidelines have been developed to simplify the complex decision-making process of authorship in scientific publications, such as those by Rennie et

al. [7] and Savitz [8, 9]. In this context, the International Committee of Medical Journal Editors (ICMJE) reached the largest application within medical journals. Nonetheless, its implementation has been inconsistent across different fields [10]. Recently, the Contributor Roles Taxonomy (CRediT) system has emerged as a promising solution to provide a more detailed account of individual contributions enhancing transparency [11]. However, even with the adoption of CRediT, challenges remain in quantifying researcher's work in a manner that equitably reflects each individual involvement.

Various instruments have been proposed over the years to objectively calculate authorship. The Authorship Scale, developed in 1997, is an instrument used in the medical field to attribute authorship. One limitation of this tool is the lack of oversight from the principal investigator (PI) to ensure the accuracy of self-reported contributions, which necessitates

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unbiased self-reporting by authors [12]. This instrument has had some adoption over the following years; nonetheless it never met a large audience of applications [13]. The Five-Step Authorship Framework, introduced in 2014, is tailored specifically for industry-sponsored clinical trial publications within the medical field. While this method helps determine whether a collaborator's contributions merit authorship, it does not provide a clear way to rank contributions against those of other collaborators, nor does it include detailed procedures for resolving conflicts or ties in rankings [14]. The Survey-weighted Analytic Hierarchy Process (S-AHP), introduced in 2021, is another tool based on the ICMJE criteria. Authorship is only warranted if a contributor has made significant input to at least one component of the ICMJE's criteria and has contributed to final approval and accountability for the study. While the S-AHP provides a structured method for determining authorship, it lacks customization for creditable criteria and levels of contribution. Additionally, it does not describe a method for resolving tied scores or other conflicts and relies on unbiased self-reporting from authors [15]. The CalculAuthor recently developed by Martins et al. introduces a method for crediting authorship by quantifying individual contributions across various project criteria. Unlike previous methods, the CalculAuthor allows for customizable criteria and weightages tailored to the specific needs of each research project. By incorporating both self-reported contributions and PI oversight, it ensures accuracy and fairness in attribution. The CalculAuthor also addresses the issue of biased self-reporting and minimizes conflicts by providing a clear conflict resolution process overseen by the PI. One of its significant strengths is its ability to promote inclusivity and equity, as it fairly recognizes contributions from all team members, regardless of seniority. This transparency and objectivity make the CalculAuthor a valuable tool for ensuring that authorship credit is fairly and accurately distributed, enhancing satisfaction and motivation among researchers [16].

The most prevalent theme regarding authorship in maxillofacial surgery and otolaryngology is undoubtedly gender equity [17]. Regarding honorary authorship (HA) - referred to authors listed without fulfilling ICMJE criteria - a recent study by Gadjradj et al. assessed its prevalence in the maxillofacial and oral surgery field. A 22-question survey was conducted among corresponding authors of articles in four high-impact journals, with a response rate of 24.8% addressing demographics, awareness of authorship guidelines, and instances of HA. Results showed that 81.1% of respondents were aware of authorship guidelines, but 56.3% were aware of HA issues, and 15.5% acknowledged having co-authors who did not meet the ICMJE criteria [18]. Notably, according to recent meta-analysis, this prevalence may be even higher reaching 20% of cases [19].

The recent advent of LLMs has significantly transformed the landscape of medical academic research [20]. These advanced generative Artificial Intelligence (AI) systems are capable of processing vast amounts of data, performing complex calculations, and generating insightful analyses, which can aid researchers in conducting systematic reviews, data synthesis with unprecedented speed and accuracy [21]. In the context of academic publishing, we hypothesise that LLMs may offer a powerful tool to objectively evaluate and compare declared contributions against actual authorship lists. By automating the assessment process, LLMs could not only enhance efficiency but also reduce potential biases, thereby promoting greater equity and accountability in scholarly authorship practices.

This manuscript addresses a critical gap in the maxillofacial surgery and otolaryngology literature by exploring the adoption of standardized authorship contribution systems integrating innovative tools like CalculAuthor and LLMs. Given the lack of extensive data on authorship equity themes in these fields, our study aimed to evaluate the adoption and application of credit systems the CRediT and ICMJE systems by assessing their prevalence within indexed journals. A secondary objective was to analyze the alignment of credit statements with actual authorship order at the manuscript level, using the CalculAuthor tool and integrating the capabilities of LLMs to enhance accuracy and objectivity in the evaluation process.

Materials and methods

Search strategy and journal selection

A search of the Scopus indexed journal was made on the 16th of March 2024; the search retrieved 838 “surgery” Journals, 777 were excluded as they were out of the topic of our study. From the remaining 61 journals, 17 were excluded because they did not have a CiteScore. The remaining 44 articles belonged to the fields of maxillofacial surgery (23 journals) and otolaryngology (21 journals) and comprised the journal study dataset.

Sampling method

A stratified random sampling method was employed to ensure representation from different quartiles. At least 2 journals from each quartile were selected, and from these, a sample of 7 articles published within the last two years was selected from each journal. The selection of articles was conducted as follows: for each journal, eligible articles published in the last two years were identified - excluding editorials, commentaries, and articles with fewer than three

authors - articles were then numbered, and a random number generator was used to select the manuscripts for analysis.

Utilization of the CRediT system, ICMJE and other contribution system

To determine the proportion of journals and articles within each field that utilized credit reporting systems, the author guidelines of each journal were reviewed. The presence of the CRediT and ICMJE system was registered as well as other author contribution guidelines.

Assessment of author contributions through large language models

In assessing the alignment between declared contributions and the effective author list for each of the 7 manuscript selected from the included journals, our methodology involved: (i) calculating the contribution scores of each author based on predefined weightages as Conceptualization (10%), Data Curation (6%), Formal Analysis (8%), Funding Acquisition (5%), Investigation (7%), Methodology (9%), Project Administration (4%), Resources (4%), Software (6%), Supervision (7%), Validation (7%), Visualization (6%), Writing - Original Draft (10%), and Writing - Review & Editing (10%); (ii) Rank authors according to their total scores; (iii) Compare the top three scored authors with the actual first, second, and last authors in the published work; (iv) Calculate the agreement percentage for each key position using the specified percentages in our formula (exact match of the first position gives 40%, any other matches gives 30%); (v) Sum these percentages to derive the total agreement score. To ensure consistent evaluation of author contributions across all sampled articles, we employed a mapping process to align contributions reported under different taxonomies with the roles defined in the CRediT Taxonomy. For journals adhering to the ICMJE guidelines or other frameworks, roles and contributions were analyzed and categorized based on their functional equivalence to the 14 CRediT roles, as previously reported [22]. By applying this systematic mapping process, we ensured consistent categorization of contributions across manuscripts, regardless of the initial taxonomy used. This approach minimized discrepancies and facilitated robust comparative analysis of authorship practices. This entire process was facilitated using ChatGPT 4-o (Open AI, San Francisco, CA, USA) through a sequence of structured prompts, which ensured that the analysis was conducted systematically and aligned with the study's objectives [23]. To validate the results, a manual human check was also performed for comparison. The specific prompts used for this analysis can be consulted in the supplementary materials.

Statistical analysis

The sample size for this study was estimated using the prevalence of honorary authorship reported in prior studies: based on a prevalence of 15.5% [18], a margin of error set at 5% and a 90% confidence level, the calculation yielded a sample size of 143 manuscripts [24]. For the primary aim, descriptive statistics summarized the number and percentage of journals adopting credit contribution systems out of the total number reviewed. A chi-square test compared the prevalence between the two research fields. For the secondary aim, descriptive statistics summarized the agreement scores across all articles. A one-sample t-test evaluated if the mean agreement score significantly deviated from a hypothesized mean agreement score of 70%, as calculated by available aggregated data on HA [19]. A correlation analysis (Spearman's rank correlation) assessed the relationship between the total contribution scores and actual authorship order. Subgroup analyses determined if there were significant differences in agreement scores based on author guidelines, journal quartiles and research fields (maxillofacial vs. otolaryngology) using Mann-Whitney U or Kruskal-Wallis tests. The study employed Jamovi (version 2.3) computer software (The Jamovi Project 2022, Sidney, Australia) for macOS Big Sur [25]. Results were considered statistically significant at a p -value < 0.05 .

Results

Out of the 21 journals analyzed, 8 adhered to the CRediT taxonomy (38.1%), 7 adhered to other contribution systems (33.3%; 4 to ICMJE and 3 to undefined systems), while 6 journals (28.6%) did not report any contribution declaration requirement in their guidelines. Out of 147 manuscripts screened, 57 (38.8%) reported contributions from 441 authors (average of 7.7 authors per manuscript), as shown in Table 1. Although credit guidelines were more often adopted by otolaryngology journals (42.9%) compared to maxillofacial surgery journals (34.8%), no statistically significant differences were found using the chi-squared test ($p = 0.143$).

The analysis of CalculAuthor across a sample of random manuscripts revealed that the mean agreement score was 65.3% (SD = 26.8). A one-sample t-test showed that the mean agreement score did not significantly deviate from the hypothesized mean of 70% ($p = 0.139$). Furthermore, the Spearman's rank correlation coefficient between the number of authors and Agreement Scores was -0.049 ($p = 0.712$) and between the CiteScore and the Agreement Score was -0.552 ($p = 0.063$). Subgroup analysis using the Kruskal-Wallis test indicated no significant differences in agreement scores

Table 1 Ranking, adoption of authorship guidelines and CalculAuthor analysis results of 12 Maxillofacial and 9 Otolaryngology journal included in the present study

Journal	CiteScore	Quartile	Guidelines	Authorship reported on a sample of 7 random manuscripts	Average Number of Authors (SD)	Average CalculAuthor Agreement Score (%)
JCMFS	5.6	1	CRedit	1	6 (NA)	40 (NA)
IJOMS	5.0	1	NR	0	NA	NA
OMSNA	4.9	1	NR	0	NA	NA
IJOMI	4.2	1	NR	0	NA	NA
OMS	2.6	2	CRedit	4	5.15 (0.96)	55 (17.32)
AMS	2.4	2	Undefined	0	NA	NA
ACS	2.3	2	ICMJE/CRedit	7	8.14 (3.39)	53.33 (25.82)
JOS	0.8	3	Undefined	3	6.07 (0.58)	67.06 (5.77)
JOMSMP	0.7	3	Undefined	2	8.5 (3.54)	100 (0)
OMS Cases	0.6	3	CRedit	7	7.14 (1.86)	68.33 (41)
FOMM	0.4	4	ICMJE	7	7.43 (2.29)	80 (24.4)
NJMS	0.4	4	NR	0	NA	NA
JAMA OHNS	10.2	1	CRedit	7	7.14 (3.98)	48.57 (30.39)
CEO	5.9	1	CRedit	6	5.43 (2.99)	60 (10.95)
ANL	3.4	2	ICMJE	5	6.29 (3.87)	85 (17.32)
EAO HND	3.2	2	CRedit	0	NA	NA
COOHNS	2.7	2	NR	0	NA	NA
COR	0.7	3	CRedit	4	4.2 (0.9)	90 (20)
OTO HNS	0.6	3	CRedit	0	NA	NA
EJENTA	0.2	4	NR	0	NA	NA
KJO HNS	0.2	4	ICMJE	7	4.0 (1)	57.14 (25.63)

Abbreviations: AMS: Annals of Maxillofacial Surgery; ANL: Auris Nasus Larynx; ACS: Archives of Craniofacial Surgery; CEO: Clinical and Experimental Otorhinolaryngology; COOHNS: Current Opinion in Otolaryngology and Head and Neck Surgery; COR: Current Otorhinolaryngology Reports; EAO HND: European Annals of Otorhinolaryngology, Head and Neck Diseases; EJENTA: Egyptian Journal of Ear, Nose, Throat and Allied Sciences; FOMM: Frontiers of Oral and Maxillofacial Medicine; IJOMS: International Journal of Oral and Maxillofacial Surgery; IJOMI: International Journal of Oral and Maxillofacial Implants; JAMA OHNS: JAMA Otolaryngology - Head and Neck Surgery; JCMFS: Journal of Cranio-Maxillo-Facial Surgery; JOMS: Oral and Maxillofacial Surgery; JOMSMP: Journal of Oral and Maxillofacial Surgery, Medicine, and Pathology; JOS: Journal of Orofacial Sciences; KJO HNS: Korean Journal of Otorhinolaryngology-Head and Neck Surgery; NA: Not Applicable; NJMS: National Journal of Maxillofacial Surgery; OMS: Oral and Maxillofacial Surgery; OMS Cases: Oral and Maxillofacial Surgery Cases; OMSNA: Oral and Maxillofacial Surgery Clinics of North America

between maxillofacial surgery (mean=62.9%) and otolaryngology journals (mean=67.9%) ($p=0.437$) or among different quartiles (mean scores: Q1=60%, Q2=60.7%, Q3=73.1%, Q4=66.4%) ($p=0.433$). Additionally, a comparison between journals following the CRedit taxonomy (mean=63.1%) and those following the ICMJE guidelines (mean=64.8%) using the Mann-Whitney U test revealed no significant difference in agreement scores between the two groups ($p=0.894$).

Comparative analysis of the agreement scores derived from the CalculAuthor tool and those obtained through manual human checks revealed no discrepancies.

Discussion

Authorship of scientific manuscript carries recognition and holds significant academic, social, and financial consequences. Moreover, it entails the duty and liability for

the material published. On this regard, the preliminary bibliometric analysis herein conducted showed insightful data: (i) Authorship contribution guidelines were highly recommended (71.4% of journals) but less adopted (38.8% of manuscripts) in maxillofacial and otolaryngology journals; (ii) CRedit represented the most employed system for authorship contribution followed by ICMJE; (iii) Overall, there was a fair agreement between credit systems declarations and effective author lists in our sample (mean agreement score 65.3%, SD=26.8) according to the CalculAuthor tool. We found non-significant differences in subgroup analysis considering journal's quartiles, authorship guidelines and research disciplines. Moreover, results show that while there appears to be a tendency for higher CiteScore to be associated with lower agreement scores, this relationship is not strong enough to be considered statistically significant based on the current sample, as shown in Fig. 1.

A survey conducted by Chambers et al. [26] to investigate trends and values in authorship assignment among 266

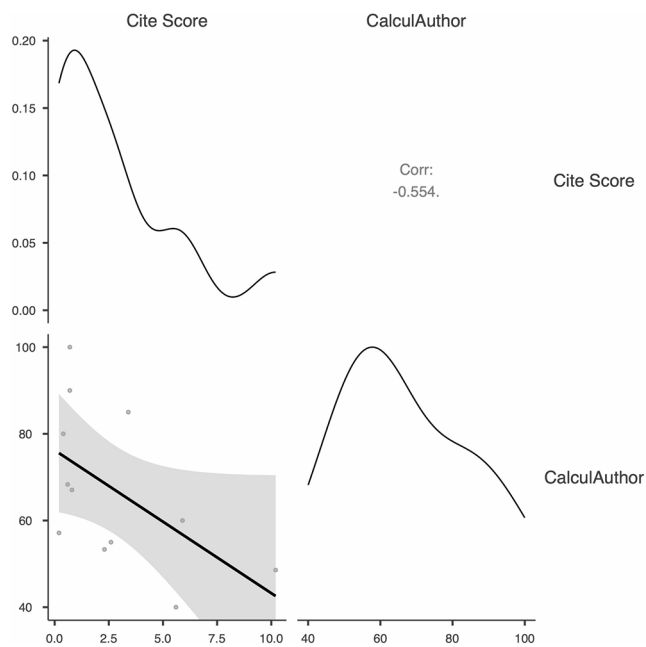


Fig. 1 Spearman correlation map between CiteScore and Authorship Agreement retrieved by CalculAuthor in the manuscripts sample of the present study, comprising Maxillofacial and Otolaryngology journals

members of the Society of Gynecologic Oncology revealed that 30.6% reported prior authorship assignment that did not meet ICMJE criteria, and 18.8% reported a history of accepting authorship not meeting these criteria. Reasons for non-adherence were including the author's patients in the study, resume building, and networking for career advancement. While the majority of respondents considered the ICMJE criteria to be generalizable and helpful, adherence to these criteria was a concern, especially among fellows and early-career faculty. A recent systematic review and meta-analysis conducted by Meursinge Reynders and colleagues assessed the prevalence of automatically listing senior members of departments as co-authors on all submitted articles in the health sciences [19]. The study reviewed survey research up to January 5, 2023, across PubMed, Lens.org, and Dimensions.ai. The analysis included 15 eligible surveys, resulting in a pooled estimate of 20% of researchers reporting this practice. Additionally, 28% of respondents felt this practice was never justified, 24% rarely justified, 25% sometimes justified, 13% most of the time justified, and 8% always justified. The findings highlight a prevalent issue where senior members are frequently added as co-authors, often without justification, indicating a need for reforms in academic authorship practices to ensure fairness and integrity in scientific publishing.

According to previous and to our results, it is desirable to spread the adoption of contribution systems guidelines in the scientific community in order to enhance transparency, equity and standardization. In this context, both CRediT

and ICMJE are efficient systems in terms of completeness and simpleness. Therefore, they should be further integrated with submission workflows as they are flexible enough to accommodate diverse article types and study designs. The application of these contribution systems reduces the ambiguity in authorship credit recognition, a flaw which is still commonly found even in high-profile maxillofacial and otolaryngology journals. For a better use of authorship credit systems, instruments such as the CalculAuthor tool [16] may be promoted. In our study we used the CalculAuthor tool to conduct a bibliometric analysis. Therefore, we proposed an ex-post application of the tool to evaluate authorship agreement scores. Our methodology, alongside the strengths of an easy and rapid application has several limitations as it only takes into account first, second and last positions in authorship orders without proper differentiation. In this context, the last author position is crucial as - in academic publishing - it traditionally denotes seniority, supervision, and intellectual leadership. As a senior researcher or principal investigator, the last author provides oversight, conceptual direction, and ensures the integrity of the research. They often contribute intellectually to the study, guide the research team, and may serve as the guarantor of the research's validity and quality. These values may be not fully represented by actual credit system calculator tools, including our proposed agreement score. Nonetheless, our results did not significantly deviate from the expected hypothesized agreement score of 70%, indicating feasibility of CalculAuthor tool to evaluate published authorship agreement in bibliometric analysis.

The present research uniquely addresses the critical need for transparency and equity in authorship attribution within the fields of maxillofacial surgery and otolaryngology (see Fig. 2). Moreover, the capabilities of LLMs to assess the alignment between declared contributions and the effective author list in selected manuscripts have been shown, providing a clear example of how these models can support and augment traditional research methodologies. While previous studies have highlighted the prevalence of HA and the challenges associated with traditional authorship guidelines, to the best of our knowledge our study is the first to systematically evaluate the adoption and implementation of the CRediT taxonomy and ICMJE guidelines across a broad range of journals in these specific fields. Additionally, the application of the CalculAuthor tool to assess the alignment of declared contributions with actual authorship order introduces a novel methodology for objectively quantifying author contributions post-publication. This approach not only provides valuable insights into current practices but also offers a potential pathway for enhancing authorship transparency and fairness, addressing a significant gap in the existing literature. By focusing on both the prevalence of credit systems and their practical impact on authorship

Authorship Transparency and Equity in Otolaryngology and Maxillofacial Surgery

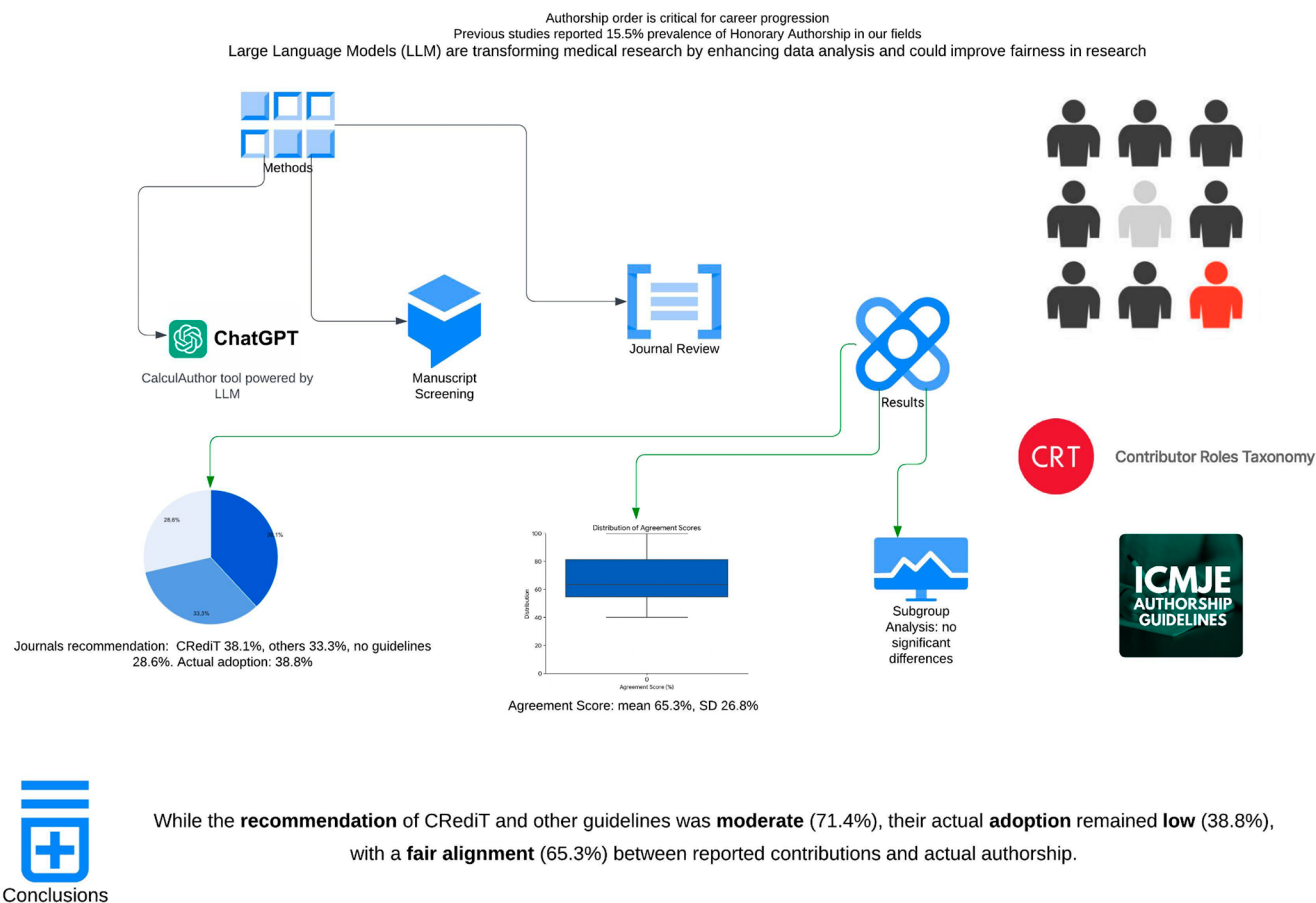


Fig. 2 Visual diagram illustrating the background, materials and methods, and results of the current study

order, our study contributes to the ongoing discourse on improving ethical standards and accountability in scientific publishing, a burden topic in the context of the generative AI era [27]. The main limitations of this study pertain to the second aim, specifically the inclusion of a small sample of manuscripts and the application of the CalculAuthor tool as a post-publication evaluation instrument, whereas it was originally designed to be used by authors before publication. Future studies should include a larger and more diverse sample of journals and manuscripts from various medical and scientific fields. Moreover, implementing tools like CalculAuthor prospectively, as part of the manuscript submission process, could enhance their adoption and ease of use and provide real-time data and insights. Leveraging advancements to develop sophisticated tools for tracking and verifying authorship contributions could revolutionize the field by providing more accurate and objective assessments, reducing the reliance on self-reporting [28]. Longitudinal studies are needed to assess the long-term impact of adopting standardized contribution systems like CRediT and ICMJE on authorship disputes, collaboration dynamics,

and researcher satisfaction and to explore the influence of these systems on career progression and recognition.

Conclusions

In conclusion, this study sheds light on the current state of authorship attribution practices in maxillofacial surgery and otolaryngology journals, revealing a moderate effective adoption of standardized contribution systems such as CRediT and ICMJE (38.8%), showing room for improvement. The use of the CalculAuthor tool powered by LLM demonstrated a fair level of agreement (65.3%) between reported contributions and actual authorship order. Our study highlights the necessity for further research and promotion of the widespread adoption of standardized authorship contribution systems. Moreover, enhancing and adopting tools like CalculAuthor will serve the scientific community, ensuring that authorship credit is accurately and equitably assigned.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00405-024-09151-5>.

Acknowledgements None.

Author contributions Conceptualization: AF; Data Curation: SB, AF; Formal Analysis: AF; Investigation: SB; Methodology: AF; Project Administration: GG, PG; Resources: SB; Supervision: GG, PG; Validation: AF, GG; Visualization: SB; Writing - Original Draft: AF; Writing - Review & Editing: SB, AF.

Funding This research received no external funding.

Data availability The original data presented in the study are openly available in <https://data.mendeley.com/> registered as frosolini, andrea (2024), “Authorship transparency and equity in maxillofacial surgery and otolaryngology: state of the art and role of the CalculAuthor tool”, Mendeley Data, V1, <https://doi.org/10.17632/ygytsrf7s.1>.

Declarations

Informed consent Not applicable.

Institutional review board Not applicable.

Conflict of interest The authors declare no conflicts of interest.

References

- Hagen NT (2008) Harmonic allocation of authorship credit: source-level correction of bibliometric bias assures accurate publication and citation analysis. *PLoS ONE* 3(12):e4021
- Fleming N (2021) The authorship rows that sour scientific collaborations. *Nature* 594(7863):459–462
- Resnik DB, Smith E (2023) Should authorship on scientific publications be treated as a right? *J Med Ethics* 49(11):776–778
- Bhattacharya S (2010) Authorship issue explained. *Indian J Plast Surg* 43(2):233–234
- Kharasch ED, Avram MJ, Bateman BT et al (2021) Authorship and publication matters: credit and credibility. *Anesthesiology* 135(1):1–8
- Thurston MM, Moniri NH, Bowen JP, Winkles CL, Miller SW (2023) Managing the Three Cs of academic literature authorship: contributions, credit, and conflict. *Am J Pharm Educ* 87(5):100009
- Rennie D, Yank V, Emanuel L (1997) When authorship fails. A proposal to make contributors accountable. *JAMA* 278:579–585
- Savitz DA (1999) What can we infer from author order in epidemiology? *Am J Epidemiol* 149:401–403
- Tarkang EE, Kweku M, Zotor FB (2017) Publication practices and responsible authorship: a review article. *J Public Health Afr* 8:723
- International Committee of Medical Journal Editors (1985) Guidelines on authorship. *Br Med J (Clin Res Ed)* 291(6497):722. <https://doi.org/10.1136/bmj.291.6497.722>
- Pulido M (2023) CRediT taxonomy: a system for identifying the contribution of each author in a scientific article. *Med Clin (Barc)* 161(2):62–64
- Ahmed SM, Maurana CA, Engle JA, Uddin DE, Glaus KD (1997) A method for assigning authorship in multiauthored publications. *Fam Med* 29(1):42–44
- Sharma H, Verma S (2018) Authorship in biomedical research: a sweet fruit of inspiration or a bitter fruit of trade. *Trop Parasitol* 8(2):62–69
- Marušić A, Hren D, Mansi B et al (2014) Five-step authorship framework to improve transparency in disclosing contributors to industry-sponsored clinical trial publications. *BMC Med* 12(1):197
- Ing EB (2021) A survey-weighted analytic hierarchy process to quantify authorship. *Adv Med Educ Pract* 12:1021–1031
- Martins RS, Mustafa MA, Fatimi AS et al (2023) The CalculAuthor: determining authorship using a simple-to-use, fair, objective, and transparent process. *BMC Res Notes* 16(1):329
- Kirubalingam K, Dzioba A, Chan Y, Graham ME (2023) Authorship diversity in otolaryngology: a 9-year analysis of articles published in *Journal of Otolaryngology-Head and Neck Surgery*. *J Otolaryngol Head Neck Surg* 52(1):57
- Gadjradj PS, Jalimsing M, Jalimsing S, Voigt I (2021) Authorship in oral and maxillofacial surgery. *J Maxillofac Oral Surg* 20(2):330–335
- Meursing Reynders RA, Cavagnetto D, Ter Riet G, Di Girolamo N, Malički M (2024) Automatically listing senior members of departments as co-authors is highly prevalent in health sciences: meta-analysis of survey research. *Sci Rep* 14(1):5883
- Frosolini A, Franz L, Benedetti S et al (2023) Assessing the accuracy of ChatGPT references in head and neck and ENT disciplines. *Eur Arch Otorhinolaryngol* 280(11):5129–5133
- Maniaci A, Saibene AM, Calvo-Henriquez C et al (2024) Is generative pre-trained transformer artificial intelligence (Chat-GPT) a reliable tool for guidelines synthesis? A preliminary evaluation for biologic CRSwNP therapy. *Eur Arch Otorhinolaryngol* 281(4):2167–2173. <https://doi.org/10.1007/s00405-024-08464-9>
- Mondal H, Mondal S, Haldar R (2023 Oct-Dec) Criteria to be an author of a manuscript: Time to revisit the ICMJE criteria and CRediT. *J Anaesthesiol Clin Pharmacol* 39(4):674–675. https://doi.org/10.4103/joacp.joacp_175_22
- ChatGPT-4o (2024) Available online: <https://openai.com/blog/chatgpt>
- Cochran WG (1977) *Sampling Techniques*. 3rd Edition. New York: Wiley
- The Jamovi Project (Version 2.3) (2024) Available online: <http://www.jamovi.org> (accessed on 28 June 2024)
- Chambers LM, Watson CH, Yao M et al (2021) Survey of trends in authorship assignment in gynecologic oncology: keeping score and playing fair. *Gynecol Oncol Rep* 36:100755
- Sallam M (2023) ChatGPT utility in healthcare education, research, and practice: systematic review on the promising perspectives and valid concerns. *Healthc (Basel)* 11(6):887
- Frosolini A, Catarzi L, Benedetti S et al (2024) The role of large language models (LLMs) in providing triage for maxillofacial trauma cases: a preliminary study. *Diagnostics (Basel)* 14(8):839

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